

R Functions Practice for Bioinformatics

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Function Exercises

In this section we will learn how to write functions by practicing.

The exercises go from **easy to difficult**.

Preparation (Test Data)

First we create some example DNA sequences.

```
dna1 <- c("A", "G", "C", "T", "G")
dna2 <- c("A", "T", "A", "T", "A", "T")
dna3 <- c("G", "G", "C", "C", "G", "G")
dna4 <- c("N", "N", "A", "T", "G")
dna5 <- c("A", "G", "X", "T") # unknown character test

dna_list <- list(dna1, dna2, dna3, dna4)
```

Level 1 — Easy (Understand how functions work)

Exercise 1: Square function

Task

Write a function that takes a number and returns the square of that number.

Function name

`square_value(x)`

Expected output

```
square_value(4)
16
```

Hint

You can calculate the result like this:

```
result <- x * x
```

Then return the result using:

```
return(result)
```

Exercise 2: DNA length

Task

Write a function that takes a DNA sequence and returns the length of the sequence.

Function name

```
dna_length(dna)
```

Expected output

```
dna_length(dna1)  
5
```

Hint

You can use the function:

```
length()
```

Level 2 — Medium (for + if + counter)

Exercise 3: Count GC bases

Task

Write a function that returns the total number of **G** and **C** bases in a DNA sequence.

Function name

```
gc_count(dna)
```

Rules

- use a **for loop**
- use an **if condition**
- use a **counter**

Expected output

```
gc_count(dna1)
3
```

Hint

Create a counter:

```
GC_total <- 0
```

Inside the loop check:

```
if (n == "G" || n == "C")
```

If true, increase the counter.

Exercise 4: GC ratio

Task

Calculate the GC ratio by dividing the GC count by the DNA length.

Function name

```
gc_ratio(dna)
```

Expected output

```
gc_ratio(c("G","C","A","T"))
0.5
```

Hint

First calculate the GC count, then do:

```
GC_total / length(dna)
```

Level 3 — Medium / Advanced (Classification)

Exercise 5: GC classification

Task

Write a function that classifies a DNA sequence based on GC ratio.

Function name

`gc_status(dna)`

Rules

- If GC ratio is greater than 0.5 → "GC-rich"
- Otherwise → "GC-poor"

Expected output

```
gc_status(dna3)
"GC-rich"
```

Hint

First calculate GC ratio, then use `if/else`.

Level 4 — Hard (Input checking)

Exercise 6: DNA validation

Task

Write a function that checks the characters in a DNA sequence.

Function name

`dna_check(dna)`

Rules

If the DNA contains a character other than **A / T / G / C / N**, return:

"ERROR: Unknown base"

Otherwise return:

"OK"

Expected output

```
dna_check(dna1)
"OK"

dna_check(dna5)
"ERROR: Unknown base"
```

Hint

Inside the loop check:

If

```
n != "A" && n != "T" && n != "G" && n != "C" && n != "N"
```

then return an error.

Exercise 7: DNA quality check (QC logic)

Task

Write a function that classifies DNA sequences using a quality control logic.

Function name

```
dna_quality(dna)
```

Rules

- If "N" exists → "Uncertain DNA"
- If GC ratio ≥ 0.6 → "High GC"
- Otherwise → "Normal DNA"

Expected output

```
dna_quality(dna4)
"Uncertain DNA"
```

Hint

First check if "N" exists.

```
N_exists <- FALSE
```

Then decide using the GC ratio.

Exercise 8: List report

Task

Write a function that takes a list of DNA sequences and prints a report for each DNA.

Function name

```
list_report(dna_list)
```

Expected behavior

For each DNA:

1. run `dna_check`
2. if there is no error run `dna_quality`
3. if there is no "N" also print GC ratio

Example output

```
list_report(dna_list)
1. DNA -> Normal DNA | GC: 0.6
2. DNA -> Normal DNA | GC: 0
3. DNA -> High GC    | GC: 1
4. DNA -> Uncertain DNA
```

Hint

You can use a counter:

```
idx <- 1
```

To combine text you can use:

```
paste()
```

Exercise 9: DNA summary function

Task

Write one function that returns a short summary of a DNA sequence.

Function name

```
dna_summary(dna)
```

Steps

1. run `dna_check(dna)`
2. if there is an error return the result immediately
3. otherwise run `dna_quality(dna)`
4. if the result is not "Uncertain DNA" also add GC ratio

Expected output

```
dna_summary(dna1)
```

```
"OK | Normal DNA | GC=0.6"
```

Hint

To combine text you can use:

```
paste()
```

Example:

```
paste("OK |", status, "| GC=", ratio)
```